

D4 peak window as a forecast export cap (0.49-c: pure function bm_d4)
goal: keep the stacks free for the afternoon peaks, charge only the export above EXPORT_CAP (summer 2026: DO4 on 55 days, 11:45-16:15)

clear-sky model dc(t) (bm_logic.c calc_arrays_kt, from fox2db_logic.h / clearsky.tex)

inputs	
time	unix clock (NTP, Fritzbox)
site	LAT 47.6811 N, LON 11.5732 E (Lenggries)
outdoor temperature	aussen/temp (MQTT, < 15 min old)
steps	
sun	NOAA: elevation, azimuth from time + site
atmosphere	air mass AM = 1 / sin(elev) (≤ 37), Meinel T = 0.7^(AM^0.678)
4 PV strings	tilt / azimuth from south / power, fitted per string (gen_pv_strings.py, 14 clear days): Sofar PV1 west 60 deg / +33 deg / 19430 W, PV2 south 68 deg / -12 deg / 7690 W FoxESS pv2 east 59 deg / -29 deg / 22036 W, pv1 west 67 deg / +32 deg / 2781 W
horizon	trees east: below 23 deg (south-east 15 deg) only 25 % diffuse
month	KT_MONTH empirical: Dec 0.13 ... May 0.91
temperature	cell = outdoor + NOCT 45 x sun; -0.30 %/K above 25 deg C (0.85..1.0)
dc(t) = Σ power x T x kt x cos(incidence) x shade x temperature factor	

dc(t)

forecast scan (ladesperre wrapper, not a decision)
now .. 21:00 in 15 min: PV = clear sky x OWM slot kt x correction,
anchored to the measured Sofar PV of now (fades over 1.5 h)
expected export = PV - HOUSE_EST 1000 W; peak_h, win_end_h (last export > cap)

bm_d4 bits	
enabled	BATMONITOR_LADESPERRE and EXPORT_CAP > 0 (18000 W)
season	May..August
fc_cap_ahead	forecast export > EXPORT_CAP before 21:00
surplus_cap_on / hold	surplus - EXPORT_CAP > 200 W / > 20 W
chg_prev	a stack charged last cycle
new_day, pcc_peak	display: PCC > 20 kW seen today

bm_d4 state: peak_today (display)
cap_active = enabled and season and fc_cap_ahead
charge above the cap = surplus_cap_on or (chg_prev and surplus_cap_hold)
block (wait, LADESPERRE 0 W) = cap_active and not charge above the cap
cap_active -> charge target PCC = +EXPORT_CAP (rule text CAP)
[make check: 384 combinations, 48 cap, 24 waiting; fixed point]

replaces 0.48-c latch / ratio / bad weather
(only 12 of 4096 combinations blocked, one cloudy
cycle ended the window for the day); the anchored
forecast follows clouds within one cycle

inputs
PCC (+ = export), PCC 5 min mean, Sofar Bat1 W (+ = charge), Multis AC-in,
outdoor temperature, BMS SoC / U / CCL, time (local, solar noon)

D5 DO4 pulse -> MQTT **pv_relay/DO4** (FoxESS shedding)
fresh PCC (< 15 s) > 20 kW -> "1", after 3 s "0", at most every 5 min
[make check]

surplus (chain_mode)
Bat1_eff = Bat1 discharge x 1, Bat1 charge x BAT1_CHARGE_FACTOR 0.5
surplus = z2 + Bat1_eff + own_charge - own_discharge
z2 = PCC + WP part the stacks do not cover (S2, as discharge)

block

D2 bm_d2 per stack state: prot, force, chg_prev
charge_mode: surplus > 200 W on, held while > 20 W
first match: BMS_MISSING > FORCE > STALE > LADESPERRE (charge_mode and block) >
CHARGE (SoC < 100) / FULL > PROT > CHARGING > DIS
[make check, CBMC h_d2_bits]

mode per stack

mode matrix BM_MODE_RULE[mode] (0.48-c): what each mode means for the setpoint

mode	action	capped	rule text
BMS_MISSING	0 W	-	NAME
FORCE	FORCE_CHARGE_W 500	1 (0.48-c)	FORCE(soc %)
STALE	0 W	-	NAME
LADESPERRE	0 W	-	LADESPERRE(peak h)
CHARGE	charge amount	1	CHARGE WP: SRC:
FULL	0 W	-	NAME
PROT	0 W	-	PROT(soc %)
CHARGING	0 W	-	CHARGING(BMS W)
DIS	discharge amount	1	(S4: PROP / IDLE)

chg_cap per phase (one per stack)
min(CHARGER_A 70 A x U / 0.93, CHARGE_MAX_PHASE 4900, BMS CCL)
~4.0-4.2 kW per phase -> ~12 kW for the three Multis

FORCE: min(FORCE_CHARGE_W, chg_cap) per phase
from the grid, obeys the BMS CCL (was: 500 W even at CCL 0)

charge amount (chain_charge, whole watts since 0.48-c)
want = max(0, own_charge + KP_PM 1010 x (z2 + Bat1_eff - target) / 1000)
target = SOYO_TARGET, + EXPORT_CAP while D4 cap_active (0.49-c)
at most the sum of chg_cap over the charging phases
[CBMC h_charge]

alloc_levels (0.48-c: one split core for charge and discharge)
charge: one level per stack in CHARGE_PRIORITY order (Stack1, Stack2), the lead stack last,
each up to min(chg_cap, CHARGER_CAP_PHASE); water filling within a level;
only what the caps cut goes on; a rest up to CHARGE_MAX_PHASE on all charging phases
[CBMC h_charge without bank assumption, h_alloc_discharge]

setpoints per phase (+ = charge) -> Multis
rule text: CHARGE WP:... SRC:... BAL/EQ [CCL] | LADESPERRE(h) | FULL | FORCE(soc)

0 W rows

clear summer day: how the export cap keeps the PCC below 20 kW

time	PV / PCC	batmonitor	stacks
morning	export below the cap	LADESPERRE (wait)	0 W, stay free
midday .. afternoon	export above 18 kW	CHARGE CAP: PCC held at 18 kW	take the export above the cap
after the forecast window	export below the cap again	CHARGE (no cap)	fill up to 100 %
cloudy forecast	no export > cap expected	no cap	charge at once
clouds come up	measured PV low	anchor drops -> cap ends	charge at once